

5 Mark(SPGD)

Q.1 What is lightning Arrestor and explain how horn Gap arrester is protecting the system against lightning?

Ans. A lightning arrester is a protective device used in electrical power systems to protect equipment such as transformers, generators, and transmission lines from high-voltage surges caused by lightning strokes or switching operations.

It is connected between the line and earth. Its main function is to divert surge current to the ground and prevent it from entering the system.

Working of Lightning Arrester:

- Under **normal operating conditions**, the arrester behaves like an **insulator** and does not conduct current.
- When a **high voltage surge** occurs, the insulation breaks down and the arrester provides a **low-resistance path to earth**.
- The surge current is discharged safely to the ground.
- After the surge passes, the arrester again becomes non-conductive.

Horn Gap Arrester

The horn gap arrester is one of the simplest types of lightning arresters used in power systems. **Construction:**

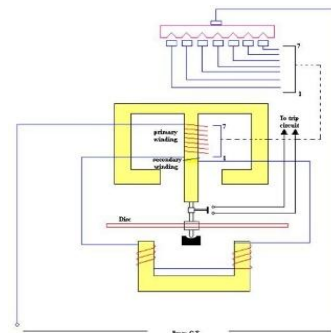
1. It consists of two horn-shaped metal conductors (electrodes) placed vertically with a small air gap between them. 2. The lower ends of the horns are close together, while the upper ends are spread apart. 3. One horn is connected to the line, and the other is connected to earth.

Working (Protection Mechanism):

- **Normal Condition:** 1. The air gap between the horns acts as an insulator. 2. No current flows. **During Lightning Surge:** 1. When a lightning surge occurs, voltage rises sharply. 2. The air gap breaks down, and an arc is formed between the horns. **Arc Movement:** 1. Due to heating, the arc becomes hot and **moves upward along the horns**. 2. The shape of the horns helps guide the arc upward. **Arc Extinction:** 1. As the arc rises, the gap between horns gradually increases. 2. The arc length increases, its resistance increases, and it finally extinguishes automatically. **Protection:** The surge current is **diverted to earth**, preventing damage to equipment.

Q. 2 Describe the construction & working principle of Induction type non directional over current relay.

Ans. Definition An induction type non-directional overcurrent relay is a protective device that operates when the current exceeds a preset value, irrespective of the direction of current flow, and trips the circuit breaker to protect the system.



Construction

1. Electromagnet (Current Coil)

- Connected to the line current through a current transformer (CT)
- Produces magnetic flux proportional to current

2. Aluminium Disc

- Mounted on a spindle
- Free to rotate between magnetic poles

3. Control Spring

- Provides restraining torque
- Sets the pickup current

4. Permanent Magnet

- Provides damping (prevents excessive rotation)

5. Moving Contact & Fixed Contact

- Close the trip circuit when the disc rotates

Working Principle

1. Normal Condition:

- Current is below the preset value
- Weak magnetic field → disc does not rotate
- Relay remains inactive

2. Fault Condition (Overcurrent):

- Current increases beyond pickup value

- Strong magnetic field induces eddy currents in the aluminium disc
- Interaction of flux and eddy currents produces torque

3. Disc Rotation:

- The disc starts rotating against the control spring
- After a certain rotation, it closes the contacts

4. Tripping Action:

- Closing of contacts energizes the trip circuit
- Circuit breaker opens and isolates the faulty section

Non-Directional Nature

- The relay operates only based on magnitude of current
- It does not depend on direction of current flow

Advantages

- Simple and reliable
- Suitable for overload and short-circuit protection
- Easy to adjust

Q. 3 • Explain types of reactor used in transmission system to limit the short circuit current.

Ans. Introduction A reactor is an inductive device used in power systems to limit short-circuit current and protect equipment like generators, transformers, and circuit breakers.

When a fault occurs, current rises sharply. Reactors introduce inductive reactance (XL) in the circuit, which reduces the fault current to a safe value.

Types of Reactors

- 1. Series Reactor**
- 2. Feeder Reactor**
- 3. Bus Bar Reactor**
- 4. Generator Reactor**
- 5. Neutral Grounding Reactor**

Advantages of Reactors

- 1. Reduce short-circuit current**
- 2. Protect equipment**

Q.4 Describe the different method of Arc Extinction of Circuit Breaker.

Ans. Methods of Arc Extinction

- **1. High Resistance Method**
- Lengthening the arc → increases resistance
- Cooling the arc → reduces ionization
- Splitting the arc into smaller arcs
- Reducing cross-sectional area of arc **Application:**
- Used in DC circuit breakers and low voltage systems

2. Low Resistance (Current Zero) Method

Working:

- When current reaches zero, arc naturally weakens
- The medium between contacts quickly regains dielectric strength
- Arc is extinguished permanently **Application:**
- Used in AC circuit breakers such as:
 - Oil circuit breaker
 - Air blast circuit breaker
 - SF₆ circuit breaker
 - Vacuum circuit breaker

. Comparison

Method	High Resistance	Low Resistance
Principle	Increase arc resistance	Use current zero
Suitable for DC systems		AC systems
Speed	Slower	Faster
Efficiency	Lower	Higher

Q.5 Describe different types of bus bar arrangement system used in substation

Ans. 1. Single Bus Bar System

- **Description:**1. Only one bus bar is used 2 All feeders are connected to it
- **Advantages:**1 Simple design 2 Low cost 3 Easy maintenance
- **Disadvantages:**1 Fault on bus → entire system shutdown 2 Low reliability

2. Double Bus Bar System

- **Description:**1 Two bus bars (Main & Auxiliary) 2 Load can be shifted between buses
- **Advantages:**1 Higher reliability 2 Maintenance possible without interruption
- **Disadvantages:**1 More costly 2 Complex operation

3. Ring Bus System

- **Description:**1 Bus bars form a closed ring 2 Each section has a circuit breaker
- **Advantages:**1 High reliability 2 Fault isolation without total shutdown
- **Disadvantages:**1 Difficult protection scheme 2 Expansion is complex

4. Breaker-and-a-Half Scheme

- **Description:**1 Uses 1.5 circuit breakers per circuit 2 Three breakers control two circuits
- **Advantages:**1 Very high reliability 2 Flexibility in operation
- **Disadvantages:**1 Very costly 2 Complex design

5. Mesh (Grid) Bus System

Description:1 Bus bars arranged in mesh form 2 Multiple paths for power flow

- **Advantages:**1 Highest reliability 2 Flexible operation
- **Disadvantages:**1 Highly complex 2 Expensive

Q.6 Write the difference between fuse and circuit breaker.

Ans. Difference Between Fuse and Circuit Breaker

Feature	Fuse	Circuit Breaker
Working principle	Melts due to heat when current exceeds limit	Opens contacts automatically using mechanical/electrical mechanism
Reusability	Cannot be reused (must be replaced)	Can be reused (just reset)
Operation speed	Very fast	Slightly slower than fuse
Cost	Low	Higher
Protection	Mainly overload & short circuit	Overload, short circuit, and advanced protection
Control	No manual control	Can be operated manually or automatically
Maintenance	Needs replacement after operation	Less maintenance, no replacement needed
Application	Used in small circuits, domestic wiring	Used in industrial and large power systems

Q.7 Write short notes on H.R.C. cartridge fuse.

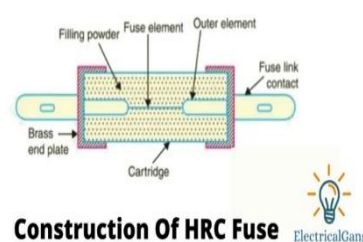
- Ans. HRC fuse is a type of fuse that can safely interrupt very high fault currents.

2 It consists of a strong ceramic body containing a silver fuse element surrounded by quartz powder (silica sand).

3 Under fault conditions, the element melts and the silica sand absorbs heat and quenches the arc.

Features:

- .High breaking capacity
- .Fast operation
- .Reliable and stable characteristics
- .Used in distribution systems and industrial circuits



Q.8 What is a voltage surge? What are the causes of over voltages?

Ans. A voltage surge is a sudden and temporary rise in voltage in a power system, lasting for a very short time (microseconds to milliseconds).

It can reach several times the normal operating voltage and may damage insulation, equipment, and appliances.

Causes of Over Voltages

1. External Causes

- **Lightning strokes:**
Direct or nearby lightning strikes induce very high voltage surges in transmission lines.
- **Electrostatic induction:**
Charged clouds can induce voltage on conductors even without a direct strike.

2. Internal Causes

- **Switching operations:**
Opening/closing of circuit breakers or switching heavy loads can create transient over voltages.
- **Faults in the system:**
Short circuits, earth faults, or sudden removal of faults can produce surges.
- **Insulation failure:**
Weak insulation can lead to abnormal voltage rise.
- **Sudden load changes:**
Rapid increase or decrease in load disturbs system voltage.
- **Resonance:**
Occurs when inductive and capacitive reactance become equal, causing excessive

In short:

- Voltage surge = sudden voltage spike
- Main causes = lightning + switching + faults + system disturbances

Q.9 Explain plain break oil circuit breakers with neat diagram.

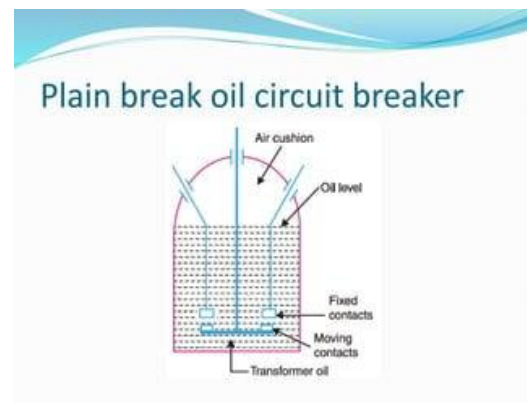
Ans. A plain break oil circuit breaker (also called bulk oil circuit breaker) is a type of circuit breaker in which oil is used as both the arc quenching medium and insulation between live parts.

Construction

It mainly consists of:

- **Steel tank** filled with insulating oil
- **Fixed contact** (stationary)
- **Moving contact** (attached to operating mechanism)
- **Insulating supports**
- **Arc chamber** (in simple designs, no special arc control device)

The contacts are completely immersed in oil.



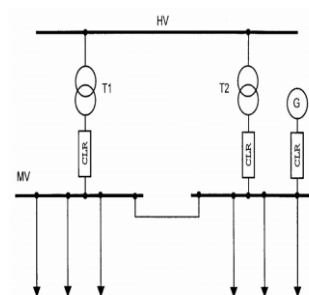
Q.10 Write down the various methods of connecting short-circuit current limiting reactors in the power systems.

Ans. Here are the standard methods used to connect short-circuit current limiting reactors in power systems:

Reactor is connected in series with each feeder

Limits fault current in that particular feeder only

Advantages: 1 Simple and effective 2 Fault isolation is easy



Disadvantages: 1 Voltage drop occurs during normal operation

2 Requires more number of reactors

2. Bus Bar Reactor Method Bus bar is divided into sections

Reactor is connected **between bus bar sections**

Advantages:

Reduces fault level across sections

Fewer reactors required

Disadvantages: Fault in one section may still affect others

Q. 11 Define (i) current rating of fuse element (ii)fusing factor (iii)cut-off current (iv)fusing current (v)breaking capacity of a fuse.

Ans. (i) Current Rating of Fuse Element

- The maximum current that a fuse can carry continuously without melting or deteriorating under normal conditions.

(ii) Fusing Factor

- It is the ratio of minimum fusing current to the rated current of the fuse.

$$\text{Fusing Factor} = \frac{\text{Minimum Fusing Current}}{\text{Rated Current}}$$

.Its value is always greater than

(iii) Cut-off Current

- The maximum value of current reached during a fault before the fuse melts and interrupts the circuit.
- It is less than the prospective fault current due to quick operation of the fuse.

- **(iv) Fusing Current**

- The minimum current at which the fuse element just melts and breaks the circuit.

(v) Breaking Capacity of a Fuse

- The maximum fault current that a fuse can safely interrupt without damage or explosion.

Q.12 Define (i) arc voltage (ii) restriking voltage (iii) recovery voltage (iv)current chopping(v)making capacity of a circuit breaker.

Ans. (i) Arc Voltage The voltage across the contacts of a circuit breaker during the arcing period (i.e., while the arc is maintained between separating contacts). **(ii) Restriking Voltage** The transient high-frequency voltage that appears across the breaker contacts immediately after arc extinction and may cause the arc to strike again. **(iii) Recovery Voltage** The normal power-frequency voltage that appears across the contacts after final arc extinction when the system returns to steady condition. **(iv) Current Chopping** The premature interruption of current by a circuit breaker before it naturally reaches zero, often causing overvoltages.

Q.13 Discuss the fundamental requirements of protective relay.

Ans. 1. Reliability

- The relay must operate correctly when a fault occurs and not fail when protection is needed.
- It should also avoid false tripping during normal conditions.

2. Selectivity (Discrimination)

- The relay should isolate only the faulty section of the system.
- Healthy parts of the system must continue operating.

3. Sensitivity

- It should be able to detect even small faults or abnormal conditions.
- Ensures protection under low fault levels.

4. Speed of Operation

- The relay must act quickly to minimize:
 - Damage to equipment
 - Disturbance to the system

5. Stability

- The relay should remain inoperative during normal conditions, overloads (within limits), and external faults.
- Prevents unnecessary tripping.

6. Simplicity

- The design should be simple and easy to operate and maintain.
- Reduces chances of errors and failures.

7. Economy

- The relay system should be cost-effective while still providing reliable protection.

Q.14 Explain low resistance method of arc extinction

Ans. Basic Idea. In this method, the arc is not forced to high resistance. Instead, its resistance is kept low, and the arc is allowed to continue until the current naturally reaches zero

Working Principle

1. When the circuit breaker contacts separate, an **arc is formed**.
2. The surrounding medium (like **oil or air**) **cools the arc**.
3. The heat reduces and **ionized particles recombine** (de-ionization).
4. As the current approaches **natural zero**, the arc becomes weak.
5. At current zero, the arc is **extinguished completely**.

Key Features

- Arc resistance is **kept low** during the process
- Relies on **natural current zero** for interruption
- Uses **cooling and de-ionization** to weaken the arc

Where It Is Used

- Oil circuit breakers
- Air circuit breakers (in some cases)

Advantages

- Simple method
- Less overvoltage compared to high resistance method

Disadvantages

- Slower compared to high resistance method
- Not suitable for very high voltage systems